King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The constant c that satisfies Rolle's Theorem for $f(x) = \dfrac{x^2 - 2x - 3}{x + 2}$ over $[-1, 3]$ is equal to:

(a) $-2 - \sqrt{5}$

(b) $1 - 2\sqrt{5}$

(c) $2\sqrt{5} - 1$

(d) $-2 + \sqrt{5}$

(e) 0

2. A water tank has the shape of an inverted circular cone with base radius $2\,m$ and height $4\,m$. If water is being pumped into the tank at a rate $2\,m^3/\text{min}$, at which rate the water level is rising when the water is $3\,m$ deep. $\left(v = \dfrac{1}{3}\pi r^2 h\right)$

(a) $\dfrac{\pi}{9}$ m/min

(b) 9π m/min

(c) 8π m/min

(d) $\dfrac{8}{9\pi}$ m/min

(e) $\dfrac{8}{3}\pi$ m/min

3. If the maximum value of $f(x) = x - \tan^{-1}(x)$ over $[0, 4]$ is A, then

(a) $A < 0$

(b) $A = 0$

(c) $|A| > 4$

(d) $A > 0$

(e) $\pi|A| < 4$

4. $x^2 + y^2 = 4$, then $\dfrac{d^2y}{dx^2} =$

(a) $\dfrac{-4}{y}$

(b) $\dfrac{x^2}{y^2}$

(c) $\dfrac{-x}{y^3}$

(d) $\dfrac{-4}{y^2}$

(e) $\dfrac{-4}{y^3}$

5. If the maximum value for $f(x) = 2\sin x - \cos 2x$ on $[0, 2\pi]$ happens at $x = a$ and the minimum value of $f(x)$ is b, then $a + b$ is equal to

(a) $1 - \pi$

(b) $\dfrac{\pi}{3}$

(c) $1 + \pi$

(d) $\dfrac{\pi - 3}{2}$

(e) $\dfrac{\pi - 1}{2}$

6. $f(x) = x^{4/5}(x - 4)^2$ has

(a) no critical numbers

(b) 3 critical numbers

(c) 4 critical numbers

(d) 1 critical number

(e) 2 critical numbers

7. One statement only is True about $f(x) = e^{\tan^{-1} x}$

(a) f is increasing on its domain

(b) The domain of f is $(-e, e)$

(c) f passes through the origin

(d) f is decreasing over $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(e) The range of $f(x)$ is $(0, \infty)$

8. $f(x) = (3-x)e^{x-3}$ is decreasing over

(a) $(2, \infty)$

(b) $(-3, \infty)$

(c) $(\sqrt{3}, \infty)$

(d) $(-\infty, \sqrt{3})$

(e) $(-\infty, 3)$

9. If $F(x) = \sqrt{g(x^2)}$, and $g(4) = 4$, $g'(4) = 3$ then $F'(2) =$

(a) $\frac{3}{2}$

(b) 4

(c) $\frac{3}{8}$

(d) $\frac{3}{4}$

(e) 3

10. $y = (\ln x)^{x^2}$, then $\left.\frac{dy}{dx}\right|_{x=e} =$

(a) e

(b) $2e$

(c) $\frac{1}{e}$

(d) $\frac{1}{e^2}$

(e) e^2

11. All edges of a cube are expanding at a rate 4 centimeters per second. How fast is the volume changing when each edge is 1 centimeter?

(a) 12 centimeter^3 per second

(b) 8 centimeter^3 per second

(c) 3 centimeter^3 per second

(d) 4 centimeter^3 per second

(e) 1 centimeter^3 per second

12. $f(x) = \sqrt{25 - x^2}$, then the slope of the tangent line at the point $(4, 3)$ is

(a) $\frac{4}{3}$

(b) $\frac{-4}{3}$

(c) $\frac{-2}{5}$

(d) 0

(e) $\frac{2}{5}$

13. Let $f(x) = \frac{x+6}{x-2}$, $x > 2$, then $(f^{-1})'(3) =$

(a) $-\frac{1}{8}$

(b) 2

(c) -2

(d) 8

(e) $-\frac{1}{2}$

14. Let $y = x\tan^{-1}(2x) - \frac{1}{4}\ln(1 + 4x^2)$, then $\frac{dy}{dx} =$

(a) $\tan^{-1}(2x)$

(b) $\tan^{-1}(2x) + \frac{1}{1+4x^2}$

(c) $\tan^{-1}(2x) - \frac{1}{1+4x^2}$

(d) $x\tan^{-1}(2x)$

(e) $\tan^{-1}\left(\frac{x}{1+4x^2}\right)$

15. $f(x) = x^{-2}\ln x$ is increasing on

(a) $\left(\frac{1}{2}e,\ e^2\right)$

(b) $(0, \sqrt{e})$

(c) (e, ∞)

(d) $(0, \infty)$

(e) $(\sqrt{e}, \infty)$

16. If $f(x) = Ax^2 + Bx + C$ is any polynomial over any closed interval $[a, b]$, then the value of c that guarantees the mean value theorem is equal to:

(a) $\dfrac{a+b}{2}$

(b) $\dfrac{a-c}{b-a}$

(c) $\dfrac{a+b+c}{b-a}$

(d) $\dfrac{b-a}{2}$

(e) $\dfrac{b^2-a^2}{a+b+c}$

17. $y = \ln|\csc x + \cot x|$, then $\dfrac{dy}{dx} =$

(a) $\csc x$

(b) $-\csc x$

(c) $\tan x$

(d) $-\sec x$

(e) $\sec x$

18. Starting with $x_1 = 2$, the second approximation x_2 to the root of the equation $x^3 - 2x - 5 = 0$ is

(a) 2.21

(b) 2.10

(c) 2.01

(d) 1.98

(e) 1.99

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$f' = \frac{x^2+4x-1}{(x+2)^2} = 0 \Rightarrow c = -2+\sqrt{5} \in (-1,3)$.
2	**(d)**	$r = \frac{h}{2}$, $V = \frac{\pi}{12}h^3$: $2 = \frac{9\pi}{4}h'$, so $h' = \frac{8}{9\pi}$ m/min.
3	**(d)**	$f' = \frac{x^2}{1+x^2} \geq 0$, so $A = 4 - \tan^{-1} 4 > 0$.
4	**(e)**	$y' = -\frac{x}{y}$, $y'' = -\frac{x^2+y^2}{y^3} = -\frac{4}{y^3}$.
5	**(d)**	Max 3 at $a = \frac{\pi}{2}$, min $b = -\frac{3}{2}$, so $a+b = \frac{\pi-3}{2}$.
6	**(b)**	$f' = \frac{2(x-4)(7x-8)}{5x^{1/5}}$: critical at $0, \frac{8}{7}, 4$, so 3.
7	**(a)**	$f' = \frac{e^{\tan^{-1} x}}{1+x^2} > 0$: f is increasing on $\mathbb{R}$.
8	**(a)**	$f' = (2-x)e^{x-3} < 0 \iff x > 2$.
9	**(e)**	$F'(x) = \frac{2x\, g'(x^2)}{2\sqrt{g(x^2)}}$, so $F'(2) = \frac{12}{4} = 3$.
10	**(a)**	Log differentiation: $\frac{y'}{y} = 2x \ln\ln x + \frac{x}{\ln x}$; at $x = e$, $y' = e$.
11	**(a)**	$V = s^3$, $V' = 3s^2 s' = 3(1)(4) = 12$ cm^3/s.
12	**(b)**	$f'(x) = \frac{-x}{\sqrt{25-x^2}}$, so $f'(4) = -\frac{4}{3}$.
13	**(c)**	$f(6) = 3$, $f'(6) = -\frac{1}{2}$, so $(f^{-1})'(3) = -2$.
14	**(a)**	The $\frac{2x}{1+4x^2}$ terms cancel, leaving $y' = \tan^{-1}(2x)$.
15	**(b)**	$f' = \frac{1-2\ln x}{x^3} > 0 \iff 0 < x < \sqrt{e}$.
16	**(a)**	$2Ac + B = A(a+b) + B$ gives $c = \frac{a+b}{2}$, the midpoint.
17	**(b)**	$y' = \frac{-\csc x(\cot x + \csc x)}{\csc x + \cot x} = -\csc x$.
18	**(b)**	$x_2 = 2 - \frac{f(2)}{f'(2)} = 2 + \frac{1}{10} = 2.10$.